



Department of Computer Science

CS4032 – Web Programming **Spring 2025**

Instructor Name: Mahhek Tahir Khan

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Office Hours: Friday 9:30 AM - 10:00AM (*email me to schedule a meeting*)

Course Information:

Program: BS(CS)/BS(DS)

Credit Hours: 3

Course Type: Elective

Course Description/Objectives/Goals:

- Introduce fundamental concepts of web architecture and both client-side (HTML, CSS, JavaScript) and server-side programming using modern frameworks.
- Apply contemporary web development practices, including AJAX, RESTful Web Services, and asynchronous programming.
- Address Web Engineering issues such as performance, scalability, security, and ethical considerations in web development.
- Foster teamwork and continuous learning through project-based development of responsive, dynamic web applications.

Program Learning Outcomes (PLOs)

This course covers the following PLOs:

PLO#	PLO Name	PLO Description
PLO 2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
PLO 3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
PLO 4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PLO 5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations
PLO 6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings

Course Learning Outcomes (CLOs):

The course learning outcomes of this course are:

CLO #	CLO Description	Domain	BT* Level	PLO #
CLO 1	Demonstrate understanding of fundamental web technologies (HTML, CSS, JS)	C	2 (Understand)	PLO 2
CLO 2	Apply web programming principles to create responsive and functional, interactive web applications	C	3 (Apply)	PLO 4
CLO 3	Analyze user requirements and web development problems to design solutions	C	4 (Analyze)	PLO 3
CLO 4	Develop dynamic, server-side or client-side applications using modern web frameworks and databases	C, P	6 (Create)	PLO 5
CLO 5	Collaborate in teams to design and develop a functional web application, adhering to best practices	C, A	5 (Evaluate)	PLO 6

BT= Bloom's Taxonomy

Domain:

C=Cognitive domain, **P=**Psychomotor domain, **A=** Affective domain

Bloom's taxonomy Levels:

1. **Remember (BT level 1):** Recall or recognize basic facts and concepts
2. **Understand (BT level 2):** Explain ideas or concepts, interpret meaning
3. **Apply (BT level 3):** Use information in new situations or apply knowledge to solve problems.
4. **Analyze (BT Level 4):** Break down information into parts to explore patterns and relationships.
5. **Evaluate (BT Level 5):** Make judgments about the value of ideas or materials based on criteria or standards.
6. **Create (BT Level 6):** Put elements together to form a new coherent whole or make an original product.

Weekly Schedule (Tentative)

Week 1	Lecture 1 Principles of Web Architecture	Lecture 2 HTTP Protocol and HTML
Week 2	Lecture 1 CSS + HTML	Lecture 2 Bootstrap, CSS Position and JavaScript
Week 3	Lecture 1 JavaScript	Lecture 2 JavaScript
Week 4	Lecture 1 JavaScript + ES6 and latest ECMAScript n versions	Lecture 2 Document Object Model
Week 5	Lecture 1 jQuery	Lecture 2 Ajax and Cookies
MID 1		
Week 6	Lecture 1 Web Architecture (MERN)	Lecture 2 System Design + Lifecycle Methods Diagram
Week 7	Lecture 1 Client-side programming (React Introduction + JSX + Routes)	Lecture 2 Client-side programming (React – Components - Class - Props)
Week 8	Lecture 1 Client-side programming (React Events – Hooks)	Lecture 2 Client-side programming (React – UI - Conditions, Form, List etc)
Week 9	Lecture 1 Client-side programming (React – Styling CSS + SASS	Lecture 2 Client-side programming (React – Redux 4.0 and Redux toolkit)
Week 10	Lecture 1 Request / Response cycle (Client-Side)	Lecture 2 Request / Response cycle (Client-Side)
MID 2		
Week 11	Lecture 1 Types of Databases and Server Side Programming (Node JS – Setup and CRUD)	Lecture 2 Server Side Programming (Node JS – File System)
Week 12	Lecture 1 Server Side Programming (Node JS – Routers)	Lecture 2 Server Side Programming (Node JS – Cookies and Session)

Week 13	Lecture 1 Server Side Programming (Node JS – Artificial Intelligence integration)	Lecture 2 MERN Full Stack - CRUD
Week 14	Lecture 1 MERN Full Stack – Authentication/Authorization	Lecture 2 Deployment – Nginx, Apache, Heroku and Dockerization

Grading Criteria:

- 📖 Assignments/Projects: 25%
- 📖 Quizzes: 10%
- 📖 Midterm Exam: 25%
- 📖 Final Exam: 40%

Course Policies:

📖 **Plagiarism** in any form (Quiz, Assignment, Midterm, and Final Exam) from any source, including the internet or other students, will result in strict penalties, which may include receiving an F grade or significant mark deductions.

📖 **Attendance Requirement:** A minimum of 80% attendance is mandatory to be eligible to appear in the Final Exam.

📖 **Passing Criteria:** To pass the course, students must achieve at least 50% overall, in accordance with the CS departments grading policies.

📖 **Grading Scheme:** An **absolute grading** system will be implemented for this course.